Characters of Sp(2m) at the powers of Kostant's element

Representation theory · characters of Lie groups 05E10 20G05 17B22 05A15

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At a glance

What the paper contributes

Kostant's element a_P is a point where every irreducible character of a compact simple group takes a value in $\{0, \pm 1\}$. The paper determines the irreducible characters of Sp(2m) at **every power** a_P^k.

- **When a character vanishes, and its value** (Theorem A): it vanishes exactly when $\lambda+\rho$ lies on more walls of the modular root arrangement than ρ does, equivalently when some folded residue class holds more entries than its capacity; otherwise the value is an explicit signed ratio of root products.
- **The absolute value is a product of dimensions** of representations of GL, Sp and odd SO, times $\kappa \in \{1, 2\}$ (Theorem B).
- **Every rational point of the same line:** a non-vanishing character is, up to sign and that factor, such a product times a character of Sp(2s) or SO(2s), $s \leq m$, at a regular element (Theorem 10.1); the points where the closed form holds for all weights are listed (Theorem C), and at the points where all characters are integers the remaining factor is 1, 2, 3, 4 or 6 (Theorem D).

Known and new, as the paper states it. For k prime to 2m+2, Theorem A is due to Cellini, Möseneder Frajria and Papi; for gcd(k, 2m+2) odd, Theorems A and B follow from Albion's factorisations of universal characters, with the sign added. New: for gcd(k, 2m+2) even and $q \geq 3$, the values on the classes not conjugate to that of ρ and the factor κ; the list of points in Theorem C and the exclusion of the others; in Theorem D, the constants for $q \geq 4$ even with $q \mid 2m$, $q \nmid m$, and for $q = 6$ with $m \equiv 4 \pmod 6$.

Key tool: an identity of residue multisets (Proposition 5.1), which takes the place of the conjugacy hypothesis in the work of Nadimpalli, Pattanayak and Prasad.

🚀 **Start here:** **Theorem A**: when the character vanishes, and its value.

Main results:

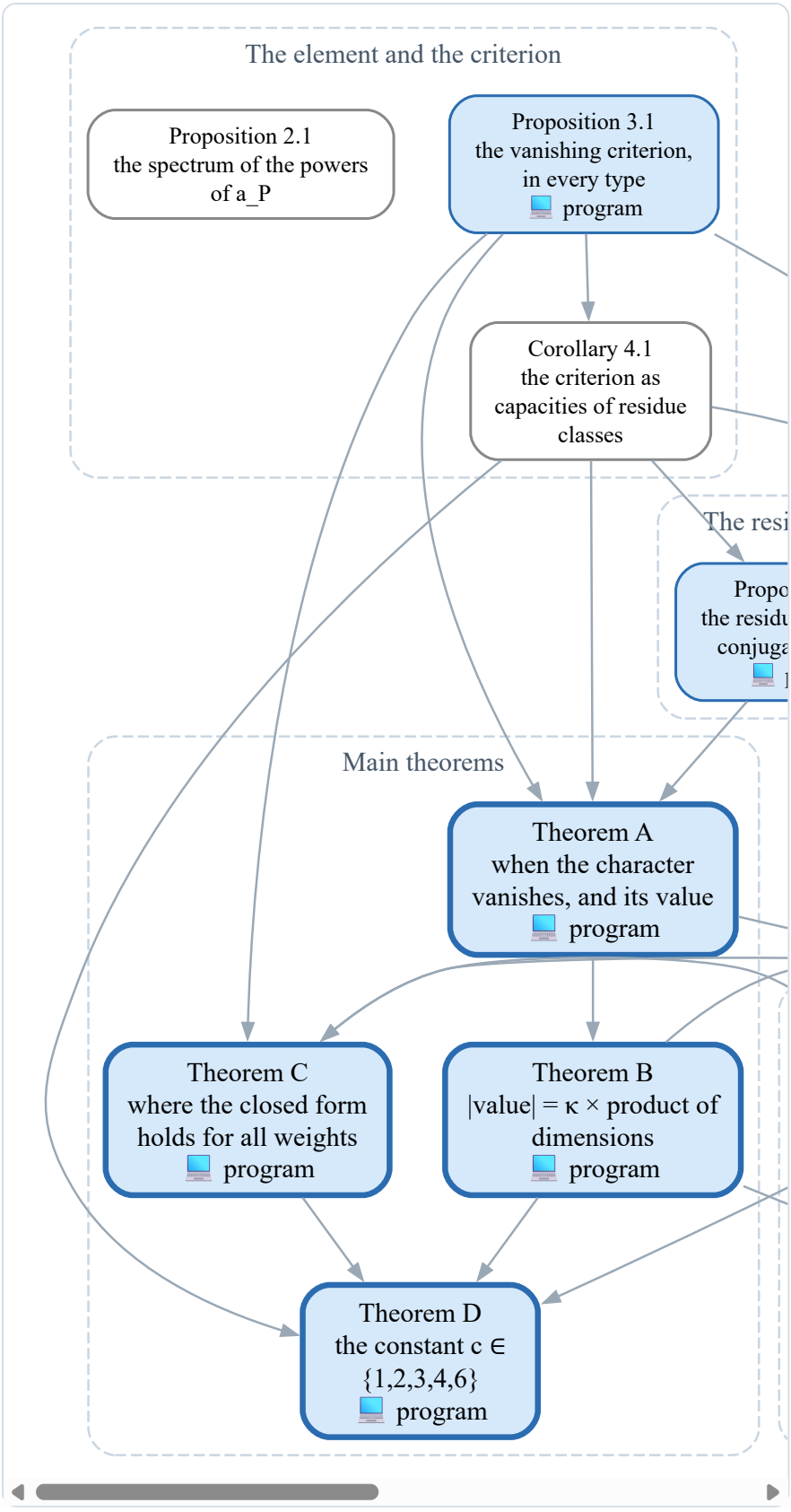
- **Theorem A**: when the character vanishes, and its value
- **Theorem B**: |value| = κ × product of dimensions
- **Theorem C**: where the closed form holds for all weights
- **Theorem D**: the constant $c \in \{1,2,3,4,6\}$

12 results | ✅ **0** formalized in Lean | ◐ **0** with a part in Lean

💻 **10** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 Setting (copied from the paper)

Let $\rho = (m, m-1, \dots, 1)$, with the form normalised so that the short roots $e_i \pm e_j$ have length 2. $g_\theta \in \mathrm{Sp}(2m)$ has eigenvalues $e^{\pm 2\pi i \theta_j}$, $j = 1, \dots, m$; $t = 2m+2$; Kostant's element is $a_P = g_{1/t}$, and $a_P^k = g_{p/q}$ with $d = \gcd(k,t)$, $q = t/d$, $p = k/d$. For $\ell = \lambda + \rho$: $N_q(x) = \#\{\alpha{>}0 : q \mid (x,\alpha)\}$, $Z_q(x) = \prod_{\alpha>0,\, q\mid(x,\alpha)} (x,\alpha)$,

$E_{p,q}(\lambda) = \sum_{\alpha>0}(\lfloor p(\ell,\alpha)/q\rfloor - \lfloor p(\rho,\alpha)/q\rfloor)$, over the positive roots $e_i \pm e_j$ $(i<j)$ and $2e_i$. Folded classes $\{c, -c\}$ modulo q; fixed if $2c \equiv 0$.

1 Main theorems

Theorem A. p. 2 map

Let λ be a dominant weight of $\mathrm{Sp}(2m)$, $k \geq 1$, and d, q, p as above.

1. *$N_q(\ell) \geq N_q(\rho)$, and $\chi_\lambda(a_P^k) \neq 0$ if and only if $N_q(\ell) = N_q(\rho)$. Equivalently, each folded class modulo q contains at most d of the entries $\ell_1, \ldots, \ell_m$, and each fixed class at most $\lfloor d/2 \rfloor$.*
2. *If $\chi_\lambda(a_P^k) \neq 0$, then*

$$\chi_\lambda(a_P^k) \;=\; (-1)^{E_{p,q}(\lambda)}\,\frac{Z_q(\lambda+\rho)}{Z_q(\rho)}.$$

🆕 **Classical and new:** for k prime to 2m+2 this is due to Cellini, Möseneder Frajria and Papi; for gcd(k, 2m+2) odd it follows from Albion's factorisations, with the sign $(-1)^E$ added. New: for gcd(k, 2m+2) even and q ≥ 3, the values on the classes not conjugate to that of ρ.

↳ Used in: Theorem B, Theorem 10.1

Proof ▼

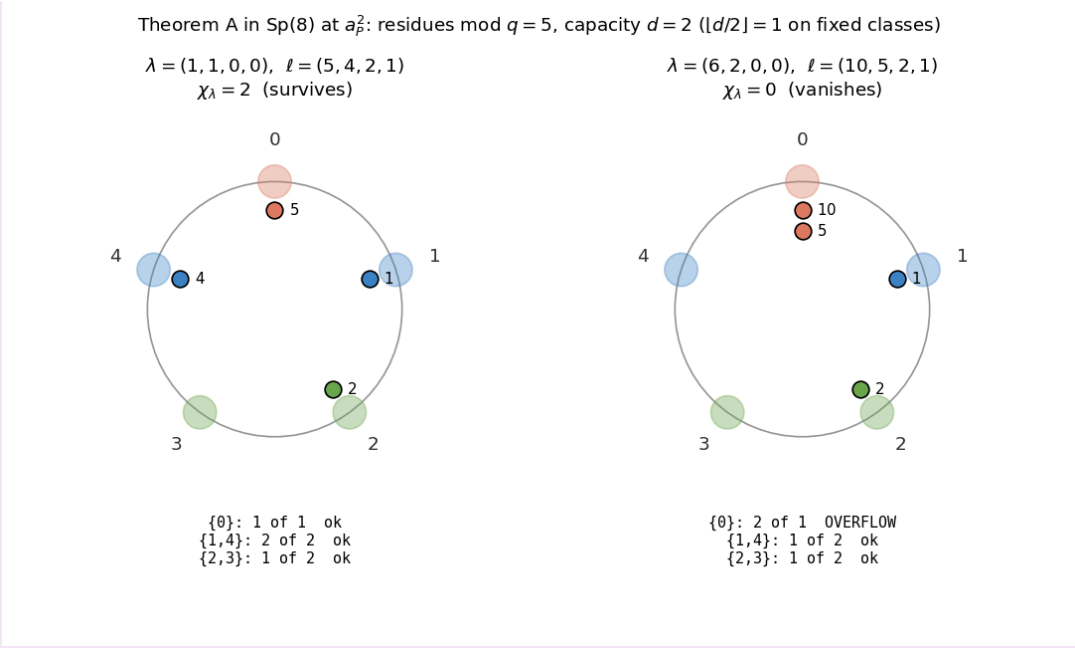
Part (1) is Proposition 3.1 together with Corollary 4.1, since $a_P^k = g_{p/q}$.

For part (2), let $\theta = p/q + \varepsilon$ with $\varepsilon \to 0$ in (3); the character is continuous in θ. For a positive integer A with $q \nmid A$ one has $\operatorname{sgn}\sin(\pi pA/q) = (-1)^{\lfloor pA/q\rfloor}$, and for $q \mid A$, $\sin(\pi A\theta) = (-1)^{pA/q}\pi A\varepsilon + O(\varepsilon^3)$. For λ dominant every (ℓ,α) and every (ρ,α) with $\alpha>0$ is a positive integer. By part (1) the numerator and the denominator of (3) carry the same number r_q of vanishing factors, so the powers of $\pi\varepsilon$ cancel, each vanishing factor leaves its exponent, and every factor leaves its sign. Hence

$$\chi_\lambda(a_P^k) = (-1)^{E_{p,q}(\lambda)}\,\frac{Z_q(\ell)}{Z_q(\rho)}\cdot\frac{\prod_{\alpha>0,\ q\nmid(\ell,\alpha)}\left|\sin\left(\pi p(\ell,\alpha)/q\right)\right|}{\prod_{\alpha>0,\ q\nmid(\rho,\alpha)}\left|\sin\left(\pi p(\rho,\alpha)/q\right)\right|}.$$

The quantity $|\sin(\pi pj/q)|$ depends only on j modulo q and up to sign, so by Proposition 5.1 and (7) the last quotient is 1.

▾ 🖼️ **Figure**



Theorem A at a glance (commentary, not part of the paper): the numbers ℓ_i placed on the residues modulo q; folded classes $\{c, -c\}$ share a colour. The character vanishes exactly when some class holds more than its capacity. Values computed exactly in $\mathbb{Z}[\zeta_q]$ by `figuras/fig_reloj_A.py` (a check in the blueprint's source, published with the blueprint).

In plain words (commentary, not part of the paper)

On the line of ρ the Weyl character formula is, up to the phase $z^{-(\lambda,\rho)}$, a product over the positive roots of $(1 - z^{(\ell,\alpha)})/(1 - z^{(\rho,\alpha)})$ (equation (2) of the paper), and at a primitive q-th root of unity a factor vanishes exactly when q divides its root value. Part (1): the character survives exactly when ℓ has no more vanishing factors than ρ; in type C this is a capacity rule on the residues of the ℓ_i (Corollary 4.1). Part (2): approaching the point along the line, each vanishing factor leaves its root value, which gives $Z_q(\ell)/Z_q(\rho)$; every factor leaves a sign, which gives $E_{p,q}$; and the remaining sines cancel, because ℓ and ρ have the same root values modulo q up to sign (Proposition 5.1). For $\gcd(k, 2m+2) = 1$ this is the type-C case of Cellini–Möseneder Frajria–Papi; for d odd the paper says that Theorems A and B are not new in substance (Albion, Ayyer–Kumari), and for q odd Reeder gives part (2) up to a constant. For d even and $q \geq 3$ the values on the classes not Weyl-conjugate to that of ρ, together with κ, are new (Example 6.1).

Evidence & scope

- **Not formalized in Lean.** Proof on paper.
- **Value and sign, checked by program `signo_Fq.py`** (supplement): 18 904 surviving weights, $m = 2, \ldots, 8$, every k with $1 \leq k < 2m+2$, $\lambda_1 \leq 9$ ($m \leq 4$), ≤ 7 ($m = 5, 6$), ≤ 5 ($m = 7, 8$); this box is in the program, not in the paper. Reference value: the character as an exact integer determinant built from the spectrum of Proposition 2.1. Exact arithmetic; value, modulus and sign each

18 904 of 18 904 (`signo_Fq_OUT.txt`). Its only negative control shows that the formula needs the survival hypothesis; there is no mutation of $E_{p,q}$.

- **Additional check, not part of the supplement:** `comprueba_traducciones_aP.py` (a check in the blueprint's source, published with the blueprint), run when this blueprint is built: 2336 cases, $m = 2, \dots, 5$, every power $k < 2m + 2$, $\lambda_1 \le 4$, character computed exactly in $\mathbb{Z}[\zeta_q]$ by the principal specialisation (equation (2), the starting point of the proof). Part (1) as the occupancy rule and part (2) as the closed form: 0 failures. Mutation (bound of the fixed classes changed to d): detected in 567 cases (`comprueba_traducciones_aP_OUT.txt`).
- **Sign form of Remark 6.2:** `valor_signo_recuento.py` V3, 11 972 surviving weights, exact, 0 failures.
- **Example 6.1:** `cifras_alco.py` E1 reproduces -2 and $+2$, $Z_3(\ell) = 6$, $Z_3(\rho) = 3$; the character is an exact polynomial in $\mathbb{Z}[z]$, evaluated at the root of unity in double precision (tolerance 10^{-9}). The individual root values and the absence of w with $w\ell \equiv \rho \pmod 3$ are not printed by any program.

Read this theorem in another mathematical language

Read this theorem in:

Commentary, not part of the paper. χ_λ is the character of the irreducible representation of $\mathrm{Sp}(2m)$ with highest weight λ, evaluated at the power $a_P^k = g_{p/q}$ of Kostant's principal element of the maximal torus. The criterion counts the positive roots α on which $\ell = \lambda + \rho$ is divisible by q: the character is nonzero exactly when ℓ lies on as few of these root hyperplanes modulo q as ρ does, and then its value is a signed ratio of the products of those root values.

Commentary, not part of the paper. Let $\zeta = e^{2\pi i/q}$. Then $\chi_\lambda(a_P^k) = sp_\lambda(z_1^{\pm 1}, \dots, z_m^{\pm 1})$, the symplectic Schur function at $z_j = \zeta^{pj}$. Put $\ell_i = \lambda_i + m - i + 1$, so $\ell_1 > \cdots > \ell_m \ge 1$, and $\rho = (m, \dots, 1)$.

- **Vanishing.** Reduce $\ell_1, \dots, \ell_m$ modulo q and identify each residue c with $-c$. The value is nonzero exactly when every class holds at most d of the ℓ_i, and the classes of 0 and (for q even) $q/2$ hold at most $\lfloor d/2 \rfloor$.
- **Value.** For $x = (x_1, \dots, x_m)$ let $Z(x)$ be the product of those numbers among $x_i - x_j$, $x_i + x_j$ $(i<j)$ and $2x_i$ that are divisible by q. If the value is nonzero, then $sp_\lambda(\dots) = (-1)^E\, Z(\ell)/Z(\rho)$, where E is the sum, over the same m^2 numbers $y(x)$, of $\lfloor p\, y(\ell)/q \rfloor - \lfloor p\, y(\rho)/q \rfloor$.

Commentary, not part of the paper. The $2m$ eigenvalues of a_P^k are the roots of unity $\zeta^{\pm pj}$, $\zeta = e^{2\pi i/q}$; as a multiset they are d copies of the q-th roots of unity with two letters removed (Proposition 2.1). A character value is a priori an element of $\mathbb{Z}[\zeta]$; at these points every character value is a rational integer (Proposition 8.1). Theorem A identifies that integer. With $\ell_i = \lambda_i + m - i + 1$, count the coincidences $\zeta^{p\ell_i} = \zeta^{\pm p\ell_j}$ $(i<j)$ and $\zeta^{p\ell_i} = \zeta^{-p\ell_i}$ (that is, $\zeta^{p\ell_i} = \pm 1$). The value is 0 exactly when there are more of them than for ρ; otherwise it is $\pm Z_q(\ell)/Z_q(\rho)$, a ratio of products of the integers

$\ell_i \pm \ell_j$ and $2\ell_i$ that are divisible by q. Those coincidences are exactly the divisibilities: $\zeta^{p\ell_i} = \zeta^{\mp p\ell_j}$ if and only if $q \mid \ell_i \pm \ell_j$, and $\zeta^{p\ell_i} = \zeta^{-p\ell_i}$ if and only if $q \mid 2\ell_i$, because p is prime to q.

Theorem B. p. 2 map

Let λ be as in Theorem A with $\chi_\lambda(a_P^k) \neq 0$. Then

$$\left|\chi_\lambda(a_P^k)\right| \;=\; \kappa(\lambda)\,\dim V_{\nu^{(0)}}^{\mathrm{Sp}_{2n_0}}\;\dim V_{\nu^{(q/2)}}^{\mathrm{SO}_{2n_{q/2}+1}}\prod_{0<c<q/2}\dim V_{\nu^{(c)}}^{\mathrm{GL}_{n_c}},$$

where n_c is the number of ℓ_i in the class c, the factor for $q/2$ is omitted when q is odd, and $\kappa(\lambda) = 1$ if d is odd or if the class 0 holds fewer than $d/2$ of the ℓ_i, and $\kappa(\lambda) = 2$ otherwise.

NEW **Classical and new:** for gcd(k, 2m+2) odd the absolute values are known (Ayyer–Kumari, Albion). New: the factor κ for gcd(k, 2m+2) even and q ≥ 3.

↳ Used in: Theorem D, Theorem 10.1, Corollary 10.2

Proof ▾

By Theorem A, $|\chi_\lambda(a_P^k)| = Z_q(\ell)/Z_q(\rho)$. The roots α with $q \mid (\ell, \alpha)$ join two indices of the same folded class, or are $2e_i$ with ℓ_i in a fixed class. We compute Z_q class by class.

A non-fixed class c with n entries. For two of its entries exactly one of $(\ell, e_i - e_j)$, $(\ell, e_i + e_j)$ is divisible by q, and its absolute value is $|\varepsilon_i\ell_i - \varepsilon_j\ell_j| = q|y_i - y_j|$; no $2e_i$ contributes. The y_i are distinct, and with $y_1 > \cdots > y_n$ the Weyl dimension formula for GL_n gives $\prod_{i<j}(y_i - y_j) = V_n \dim V_{\nu^{(c)}}^{\mathrm{GL}_n}$, where $V_n = \prod_{j<n} j!$. The contribution is $q^{\binom{n}{2}} V_n \dim V_{\nu^{(c)}}^{\mathrm{GL}_n}$.

The class 0 with n entries $\ell_i = qz_i$, $z_1 > \cdots > z_n \geq 1$. All of $e_i \pm e_j$ and $2e_i$ contribute, with values $q(z_i \pm z_j)$ and $2qz_i$. Since $z = \nu^{(0)} + \rho_{C_n}$, the Weyl dimension formula for Sp_{2n} gives the contribution $q^{n^2} C_n \dim V_{\nu^{(0)}}^{\mathrm{Sp}_{2n}}$, where $C_n = 2^n \prod_{i<j}\big((n-i+1)^2 - (n-j+1)^2\big)\, n! = 2^n \prod_{j\leq n}(2j-1)!$.

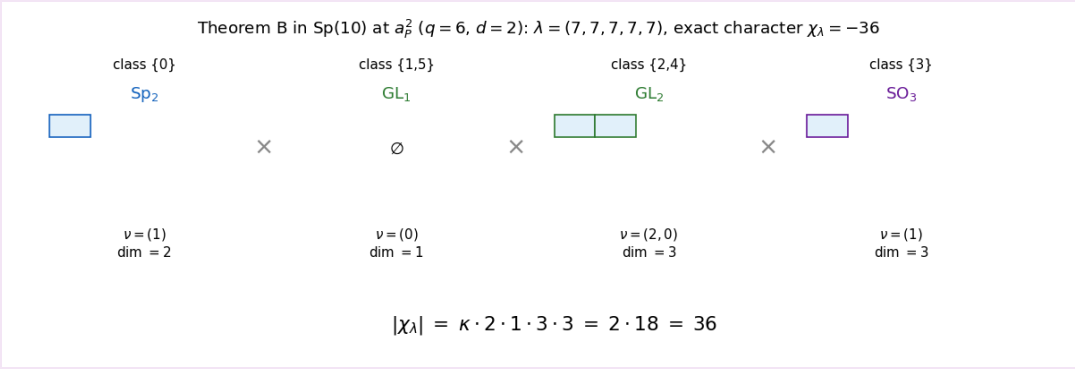
The class $q/2$ with n entries $\ell_i = qz_i$, $z_i \in \frac{1}{2} + \mathbb{Z}$. Again all of $e_i \pm e_j$ and $2e_i$ contribute, and since $z = \nu^{(q/2)} + \rho_{B_n}$ with $\rho_{B_n} = (n - \frac{1}{2}, \ldots, \frac{1}{2})$, the Weyl dimension formula for SO_{2n+1} gives $q^{n^2} B_n \dim V_{\nu^{(q/2)}}^{\mathrm{SO}_{2n+1}}$, where $B_n = \prod_{j\leq n}(2j-1)!$.

The same computation applies to ρ, whose entries in each class form an unbroken run (the class c receives $c, q+c, 2q+c, \ldots$ and $q-c, 2q-c, \ldots$, so the y_i are consecutive integers; the class 0 receives $q, 2q, \ldots$; the class $q/2$ receives $q/2, 3q/2, \ldots$), so all its partitions are empty and all its dimensions are 1. The powers of q agree on the two sides, since $N_q(\ell) = N_q(\rho)$. Hence $Z_q(\ell)/Z_q(\rho)$ is the product of the dimensions in the statement times the ratio of the constants V_{n_c}, C_{n_0}, $B_{n_{q/2}}$ for the occupancies of ℓ and of ρ. If d is odd the occupancies agree (Section 4) and the ratio is 1. If d is even, put $h = d/2$; the class 0 of ρ holds $h-1$ entries and every other class is full; if the class of ℓ that falls short is 0, the occupancies agree again; otherwise that class holds one entry less and the class 0 one entry more than for ρ, and the ratio is

$$\frac{C_h/C_{h-1}}{V_{2h}/V_{2h-1}} = \frac{2(2h-1)!}{(2h-1)!} = 2 \qquad \text{or} \qquad \frac{C_h/C_{h-1}}{B_h/B_{h-1}} = 2$$

according as that class is non-fixed or is $q/2$.

🖼️ Figure



Theorem B at a glance (commentary, not part of the paper): each folded class becomes a factor (Sp for the class 0, SO for $q/2$, GL otherwise) with its partition $\nu^{(c)}$; the absolute value of the character is κ times the product of their dimensions. The example is chosen by the script, which computes κ by the rule of the theorem ($\kappa = 1$ if d is odd or the class 0 holds fewer than $d/2$ entries, else 2), not as $|\chi|/\Delta$, and checks $|\chi| = \kappa\,\Delta$ against the exact character in $\mathbb{Z}[\zeta_q]$ on all 1232 weights it examines (`figuras/fig_factorizacion_B.py`, a check in the blueprint's source, published with the blueprint).

💡 In plain words (commentary, not part of the paper)

By Theorem A, $|\chi_\lambda(a_P^k)| = Z_q(\ell)/Z_q(\rho)$, and the proof computes Z_q one folded class at a time. A root enters Z_q only if it joins two entries of the same class, or is $2e_i$ with ℓ_i in a fixed class. In a non-fixed class the values that enter are q times differences of the y_i: the Weyl dimension formula of GL_n. In the class 0 they are $q(z_i \pm z_j)$ and $2qz_i$: the formula of Sp_{2n}. In the class $q/2$, with half-integer z_i: the formula of SO_{2n+1}. For ρ every partition is empty. What is left is a ratio of normalising constants, which is 1 unless d is even and the class that falls short is not 0; then it is 2, and that is κ. The groups are not centralisers: at $m = 6$, $k = 7$ the factorisation group is $\mathrm{Sp}_6 \times \mathrm{SO}_7$, the centraliser $\mathrm{Sp}_6 \times \mathrm{Sp}_6$ (Question 12.1).

🛡️ Evidence & scope

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `dims_recuento.py`: 11 972 surviving weights, $m = 2, \ldots, 7$, every k, $\lambda_1 \le 8$ ($m \le 4$), ≤ 6 ($m \ge 5$), box not stated in the paper. Exact (rational arithmetic, integer determinant); 11 972 of 11 972 (`dims_recuento_OUT.txt`). Like `signo_Fq.py`, it builds the spectrum from Proposition 2.1. It has no negative control for κ: its column "kappa=1 mal" is not a

mutation. A $\kappa \equiv 1$ mutation exists in `residual_factorizacion.py`, in the setting of Theorem 10.1 (floating point; breaks 371 of 1145).

- **Example 7.1:** `cifras_alco.py` E2, $Z_3(\ell)/Z_3(\rho) = 419904/52488 = 8$ and $E = 17$ exactly, value -8 in double precision (tolerance 10^{-9}). The partitions and the dimensions 4 and 2 are not printed.
- **End of Section 7:** `certifica_testigos.py`, $\chi_{(1,1,0,0,0,0)}(a_P^7) = -7$ exactly; `cifras_alco.py` E5, $d = 7$, $q = 2$, eigenvalues ± 1 six times each. The traces 0 and -6 and the absence of a 7-dimensional irreducible of $\mathrm{Sp}_6 \times \mathrm{Sp}_6$ are not checked by any program.
- **No program:** $\chi_{jq\rho}(a_P^k) = (jq+1)^{r_q}$ (introduction), and the constants C_n, B_n, V_n with the ratio 2 (proof on paper).

▾ 🔤 **Read this theorem in another mathematical language**

Read this theorem in:

Commentary, not part of the paper. The absolute value of the character is $\kappa(\lambda) \in \{1, 2\}$ times the dimension of an irreducible representation of the product group $\mathrm{Sp}_{2n_0} \times \mathrm{SO}_{2n_{q/2}+1} \times \prod_{0<c<q/2} \mathrm{GL}_{n_c}$, one factor for each folded class c holding n_c of the entries of $\ell = \lambda + \rho$, with highest weights the partitions $\nu^{(c)}$. This group is not the centraliser of a_P^k (the paper gives an example), and Question 12.1 asks for a group that explains the product.

Commentary, not part of the paper. With $\ell_i = \lambda_i + m - i + 1$ grouped by folded residue class modulo q:

$$|sp_\lambda(z^{\pm 1})| = \kappa \cdot sp_{\nu^{(0)}}(1^{n_0}) \cdot so_{\nu^{(q/2)}}(1^{n_{q/2}}) \cdot \prod_{0<c<q/2} s_{\nu^{(c)}}(1^{n_c}),$$

a product of principal specialisations at 1 of smaller symplectic, odd orthogonal and Schur functions. For a class $0<c<q/2$: choose signs with $\varepsilon_i \ell_i \equiv c$, list $y_i = (\varepsilon_i \ell_i - c)/q$ decreasingly and put $\nu_i^{(c)} = y_i - y_n - (n-i)$; then $s_{\nu^{(c)}}(1^{n_c})$ is the number of semistandard Young tableaux of shape $\nu^{(c)}$ with entries at most n_c. For the class 0, $\nu_i^{(0)} = \ell_i/q - (n_0 - i + 1)$; for $q/2$, $\nu_i^{(q/2)} = \ell_i/q - (n_{q/2} - i + \frac{1}{2})$.

Commentary, not part of the paper. The character value is a rational integer (Proposition 8.1), and Theorem B factors its absolute value: $\kappa \in \{1, 2\}$ times a product of integers, one per residue class of the ℓ_i modulo q up to sign. Each factor is a Weyl dimension, an explicit product of ratios of integers built from the quotients ℓ_i/q (classes 0 and $q/2$) or from the shifted quotients $(\pm\ell_i - c)/q$ (other classes). So the size of the value is read off from how the ℓ_i distribute among the residues modulo q.

🧪 The lab under Theorem A also shows this factorisation: for a surviving character it lists the factor of each class (Sp, SO or GL, its partition and dimension), κ, and checks the product against the exact character.

Theorem C. 📕 p. 3 📍 map

Let $m \geq 2$ and $q \geq 2$. The identity

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)}\,\frac{Z_q(\lambda+\rho)}{Z_q(\rho)}$$

holds for every dominant weight λ with $N_q(\lambda+\rho) = N_q(\rho)$ and every p prime to q if and only if $q \mid m$, $q \mid 2m+1$, $q \mid 2m+2$, or $q = 6$ and $m \equiv 1 \pmod 6$. Otherwise there is such a λ for which it fails at $p = 1$ even up to sign.

🆕 **Classical and new:** new: the list of points, and the exclusion of all the others.

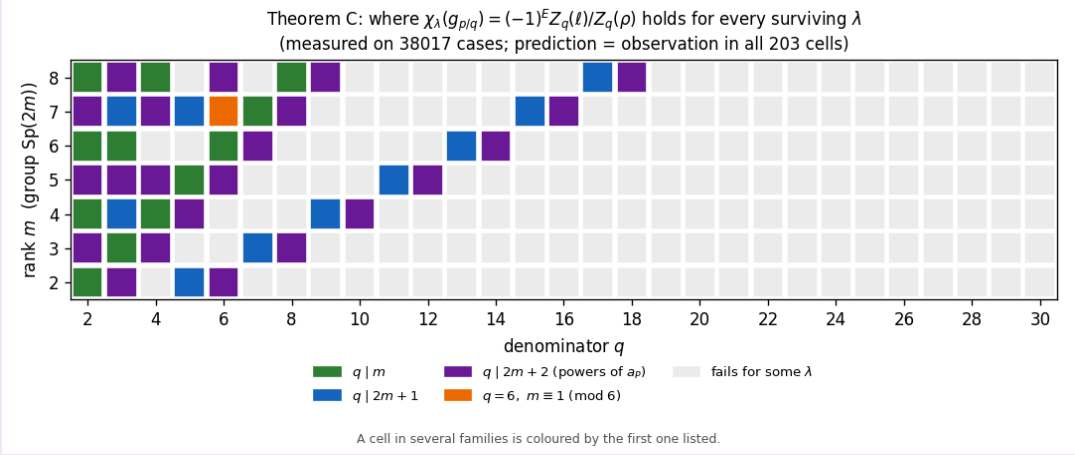
↳ Used in: Theorem D

Proof ▼

The four families give the identity. If $q \mid m$, $q \mid 2m+1$ or $q \mid 2m+2$, then $v \in \{0, a, q-1\}$ and $R_p(b) = 1$ by Theorem 10.1. If $q = 6$ and $m \equiv 1 \pmod 6$, then $v = 1 \leq a = 2$, $Y = C_1$, $\rho_Y = (1)$ and $b = (b_1)$ with $b_1 \in \{1, 2\}$, so by (9) $R_p(b) = |\sin(\pi p b_1/3)|/|\sin(\pi p/3)| = 1$, since $p \equiv \pm 1 \pmod 6$.

No other point does. Suppose that the identity holds up to sign at $p = 1$ for every surviving λ. Then every $\chi_\lambda(g_{1/q})$ is rational, the non-surviving ones being 0 by Proposition 3.1, and an algebraic integer, hence an integer; by Proposition 8.1, q divides $2m$, $2m+1$ or $2m+2$, or $q = 6$ and $m \equiv 1 \pmod 3$. The cases not listed in Theorem C are (i) $q \geq 4$ even, $q \mid 2m$ and $q \nmid m$, that is $m \equiv q/2 \pmod q$, and (ii) $q = 6$ and $m \equiv 4 \pmod 6$; for $q = 2$ either $q \mid m$ or $q \mid 2m+2$. In both we take the natural representation, $\lambda = (1, 0, \ldots, 0)$, and $\xi = e^{2\pi i/q}$. Its character is $\sum_{j=1}^{m}(\xi^j + \xi^{-j})$, and full turns contribute 0, so it equals $\sum_{j=1}^{v}(\xi^j + \xi^{-j})$. In case (i), $v = q/2$ and this is $\sum_{j=1}^{q-1}\xi^j + \xi^{q/2} = -2$; in case (ii), $v = 4$ and it is $2\sum_{j=1}^{5}\xi^j - (\xi^5 + \xi^{-5}) = -2 - 1 = -3$. In particular λ is surviving. On the other hand $\ell = \rho + e_1$, the roots not involving the index 1 take the same values on ℓ and on ρ, and those involving it take the values $\{1, \ldots, 2m\} \smallsetminus \{m\}$ on ρ and $\big(\{2, \ldots, 2m\} \smallsetminus \{m+1\}\big) \cup \{2m+2\}$ on ℓ. None of 1, m, $m+1$, $2m+2$ is divisible by q: in case (i) they are congruent to 1, $q/2$, $q/2+1$, 2, and in case (ii) to 1, 4, 5, 4. So the two sides have the same multiples of q, $Z_q(\ell) = Z_q(\rho)$, and the identity would give ± 1. The failure holds at every p, not only at $p = 1$: since $\chi_\lambda(g_{p/q}) = \sigma_p\big(\chi_\lambda(g_{1/q})\big)$ for the automorphism σ_p of $\mathbb{Q}(\xi_q)$, a rational value at p forces a rational value at 1, and the natural representation gives -2, respectively -3, at every p.

▾ 🖼️ **Figure**



Theorem C at a glance (commentary, not part of the paper): each cell (q, m) is coloured when the closed form holds for every surviving weight of the program's box ($\lambda_1 \leq 30, 16, 11, 8, 6, 5, 5$ for $m = 2, \ldots, 8$), by the family that explains it (a cell in several families takes the first one in the legend). Measured, not drawn from the statement: the colours come from the archived output of `formula_pura.py` (203 cells; 38017 cases, counted up to the first counterexample in the grey cells), and the script checks that prediction and observation agree in every cell (`figuras/fig_mapa_C.py`, a check in the blueprint's source, published with the blueprint).

💡 In plain words (commentary, not part of the paper)

The closed form of Theorem A(2) can be written at any rational point $g_{p/q}$. Theorem C says where it is right for every surviving weight and every p: $q \mid m$, $q \mid 2m+1$, $q \mid 2m+2$ (the powers of a_P), or $q = 6$ with $m \equiv 1 \pmod 6$. By equation (8) of the paper the true value is the closed form times a residual factor $R_p(b)$, so the question is when $R_p(b) = 1$; in the four families it is. Conversely, if the closed form held up to sign at $p = 1$, every character at $g_{1/q}$ would be an integer, so q is one of the points of Proposition 8.1. That leaves two integer families outside the list, $q \geq 4$ even with $q \mid 2m$, $q \nmid m$, and $q = 6$ with $m \equiv 4 \pmod 6$; there the natural representation has value -2, resp. -3, while the closed form gives ± 1. The paper states that the list and the exclusion are new.

🛡️ Evidence & scope

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `formula_pura.py`: the 203 pairs (m, q), $m = 2, \ldots, 8$, $2 \leq q \leq 30$, characters computed exactly in $\mathbb{Z}[X]/(\Phi_q)$. In every pair the prediction (the pair is in the list) equals the observation (the closed form holds on all surviving (λ, p) of the box); discrepancies: none (`formula_pura_OUT.txt`). Box $\lambda_1 \leq 30, 16, 11, 8, 6, 5, 5$ for $m = 2, \ldots, 8$, in the program, not in the paper. The 38 017 cases are counted up to the first counterexample in each pair outside the list, not over the whole box.

- **Natural representation** in cases (i) and (ii) of the proof, 56 pairs with $m \le 30$: survives, Z-quotient 1, value -2, resp. -3; exact. For the other pairs outside the list the program prints a counterexample but does not test the clause "fails even up to sign".
- **One exact implementation:** the principal specialisation in `evlib.py`, which is equation (2), the starting point of the proof. A second, floating-point evaluator (`carac.py`) confirms the closed form only in the families $q \mid m$ and $q \mid 2m+1$, on the 312 cases of `valor_todo_q.py casos`; the converse and the family $q = 6$, $m \equiv 1 \pmod 6$ have no second evaluator.
- **Section 8 examples:** `cifras_alco.py` E4, $\mathrm{Sp}(4)$, $\lambda = (1,0)$: -2 at $q = 4$ and $0.8019377358 = 2\cos\frac{2\pi}{7} + 2\cos\frac{4\pi}{7}$ at $q = 7$, Z-quotient 1 in both; double precision. The value $\chi_{(7,7)}(g_{1/7}) = 1$ is not checked by any program.

Theorem D. p. 3 map

Let $m \ge 2$, let $q \ge 2$ be one of the denominators of Proposition 8.1, and let λ be dominant with $\chi_\lambda(g_{p/q}) \ne 0$. Then

$$\left|\chi_\lambda(g_{p/q})\right| = c_\lambda\, \Delta_\lambda, \qquad c_\lambda \in \{1,2,3,4,6\}.$$

More precisely, $c_\lambda = 1$ if $q \mid m$, $q \mid 2m+1$, or $q = 6$ and $m \equiv 1 \pmod 6$; $c_\lambda = \kappa(\lambda) \in \{1,2\}$ if $q \mid 2m+2$; $c_\lambda \in \{1,2\}$ if $q \ge 4$ is even, $q \mid 2m$ and $q \nmid m$; and $c_\lambda \in \{1,2,3,4,6\}$ if $q = 6$ and $m \equiv 4 \pmod 6$. In the last two cases every listed value occurs.

NEW **Classical and new:** the families q | m and q | 2m+1, and q | 2m+2 with gcd(k, 2m+2) odd, are known through the references of the paper (Ayyer–Kumari, Albion). New: κ for gcd(k, 2m+2) even and q ≥ 3, and the constants for q ≥ 4 even with q | 2m, q ∤ m, and for q = 6 with m ≡ 4 (mod 6).

↳ Not used by another result of the paper.

Proof ▼

By Proposition 8.1, $g_{p/q}$ is conjugate to $g_{1/q}$, so in (8) the value $|\chi_\lambda|$ and the factor $\kappa\Delta_\lambda$ do not depend on p, and $c_\lambda = \kappa\, R_1(b)$. If $q \mid m$, $q \mid 2m+1$, or $q = 6$ and $m \equiv 1 \pmod 6$, then $v \le a$, so $\kappa = 1$, and $R_1(b) = 1$ by the proof of Theorem C. If $q \mid 2m+2$ this is Theorem B. Two cases remain.

$q \ge 4$ even, $q \mid 2m$, $q \nmid m$. Put $h = q/2$. Then $v = h > a$, $s = q - v = h$ and $Y = D_h$. The folded classes are $0, 1, \dots, h$, and exactly one of them, say j, holds its full number of entries; b lists the other h, while $\rho_Y = (h-1, \dots, 0)$ omits $j = h$. With $p = 1$ the numbers $t_k = \cos(\pi k/h)$, $0 \le k \le h$, are distinct, and with $\omega(x) = \prod_{k=0}^{h}(x - t_k)$ the product of $|t_i - t_k|$ over the pairs not containing j is the product over all pairs divided by $|\omega'(t_j)|$. By (9),

$$R_1(b) = \frac{|\omega'(t_h)|}{|\omega'(t_j)|}.$$

The roots of ω are ± 1 and the zeros of the Chebyshev polynomial U_{h-1}, so ω is a constant multiple of $(x^2-1)U_{h-1}(x)$, which equals $-\sin\theta\,\sin h\theta$ at $x=\cos\theta$. At an interior node $\theta=\pi k/h$, $0<k<h$, its derivative in x has absolute value h; at $x=\pm 1$ it is $\pm 2U_{h-1}(\pm 1)$, of absolute value $2h$, since $U_{h-1}(\pm 1)=(\pm 1)^{h-1}h$. Hence $R_1(b)=1$ if $j\in\{0,h\}$ and $R_1(b)=2$ otherwise. Since $\kappa=2$ exactly when the class 0 is full, that is $j=0$, one gets $c_\lambda=1$ for $j=h$ and $c_\lambda=2$ for every other j.

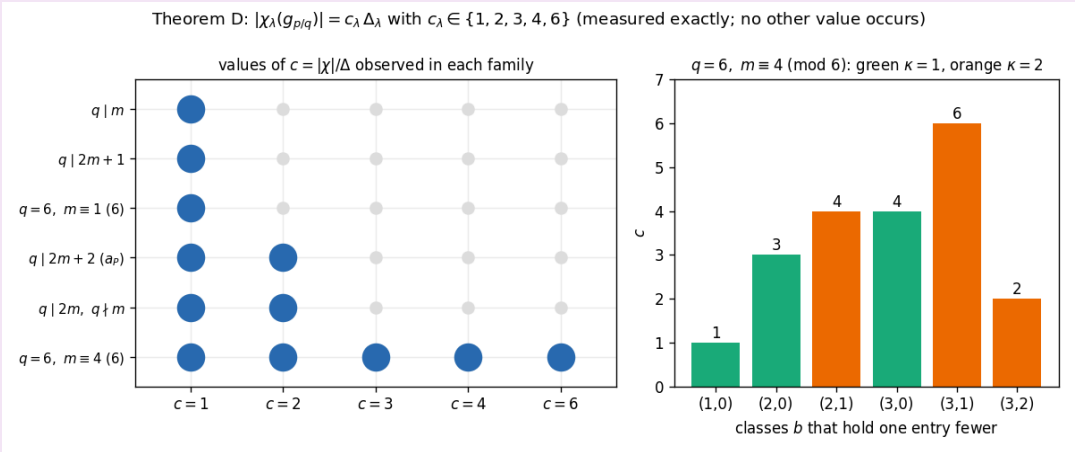
$q=6$, $m\equiv 4 \pmod 6$. Here $v=4>a=2$, $s=2$, $Y=D_2$ and $\rho_Y=(1,0)$; two of the classes $0,1,2,3$ fall one entry short, and $b=(b_1,b_2)$ lists them. With $p=1$, (9) gives $R_1(b)=2\,|\cos(\pi b_1/3)-\cos(\pi b_2/3)|$, and $\kappa=2$ exactly when $0\notin b$:

b	$(1,0)$	$(2,0)$	$(3,0)$	$(2,1)$	$(3,1)$	$(3,2)$
$R_1(b)$	1	3	4	2	3	1
κ	1	1	1	2	2	2
c_λ	1	3	4	4	6	2
λ $(m=4)$	0	(1)	$(1,1)$	(2)	$(2,1)$	$(2,2)$
$\chi_\lambda(g_{1/6})$	1	-3	4	4	-6	2

The last two rows give, for $m=4$, a weight realising each column, with $\Delta_\lambda=1$; the values are computed exactly in $\mathbb{Z}[\zeta_6]$ (`certifica_testigos.py`).

In the last two cases every occupancy pattern of a surviving weight, that is every choice of the classes that fall short, is the pattern of some $\lambda+\rho$, by the argument in the proof of Corollary 4.1; so every listed value of c_λ occurs.

Figure



Theorem D at a glance (commentary, not part of the paper): left, the values of $c=|\chi|/\Delta$ observed in each family, for $m\in\{2,\dots,8,10,13,16\}$ with a box on λ per m; the points that lie in two families ($q\mid 2m+2$ together with $q\mid m$, or with $q\mid 2m$, $q\nmid m$) are not shown; right, the exceptional case $q=6$, $m\equiv 4 \pmod 6$, where c depends on which two classes hold one entry fewer. Data from the archived output of `prefactores.py`, exact fractions; the script checks it against the table as the program prints it, transcribed from the paper (`figuras/fig_constantes_D.py`, a check in the blueprint's source, published with the blueprint).

▾ 💡 **In plain words (commentary, not part of the paper)**

At the integer points of Proposition 8.1 every non-vanishing character is c_λ times the product of dimensions Δ_λ of Theorem B, with $c_\lambda \in \{1,2,3,4,6\}$. Since $g_{p/q}$ is conjugate to $g_{1/q}$ there, $c_\lambda = \kappa\, R_1(b)$. In the families of Theorem C other than $q \mid 2m+2$, $\kappa = 1$ and $R_1(b) = 1$; for $q \mid 2m+2$ it is Theorem B. Two cases are left. For $q \geq 4$ even, $q \mid 2m$, $q \nmid m$, the residual group is $D_{q/2}$ and $R_1(b)$ is a ratio of derivatives of $(x^2-1)U_{h-1}(x)$ at the nodes $\cos(\pi k/h)$, $h = q/2$: 1 at the end nodes, 2 inside. So $c_\lambda = 1$ when the full class is $q/2$, and 2 otherwise. For $q = 6$, $m \equiv 4 \pmod 6$, the group is D_2 and the six choices of the two short classes give the table of the proof: $c_\lambda = 1, 3, 4, 4, 6, 2$.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `prefactores.py`: integrality at the points of Proposition 8.1 on 52 928 pairs (λ, q), exact in $\mathbb{Z}[X]/(\Phi_q)$, 0 non-integers; the set of $c = |\chi|/\Delta$ in each family, as exact fractions, equal to the statement, with every listed value occurring in the last two families; 0 values outside $\{1,2,3,4,6\}$ (`prefactores_OUT.txt`). Range $m \in \{2,\dots,8,10,13,16\}$ with a box on λ per m, in the program, not in the paper.
- **The table:** same program, (κ, c) for each pair b, printed next to the row c_λ as transcribed in the program, and equal to it; the end-node/interior rule, (κ, c) by omitted node, for the even $q \leq 32$ that occur in its range (the end node 0 only for $q \leq 12$). Both are printed, not asserted. `certifica_testigos.py`: the last two rows ($m = 4$, values $1, -3, 4, 4, -6, 2$, $\Delta_\lambda = 1$) exactly in $\mathbb{Z}[\zeta_6]$. The map from λ to b in the table is not checked by any program.
- **Partial second implementation:** `teorema_II.py` (b) and `residual_factorizacion.py` (V2, V3) find the same set $\{1,2,3,4,6\}$ and the six rows of the table, in floating point, on a smaller range.
- No negative control in `prefactores.py`.

2 The element and the criterion

Proposition 2.1. p. 5 map

> *Let $t = 2m+2$, $d = \gcd(k,t)$ and $q = t/d$. As a multiset, the spectrum $A = \{\zeta^{\pm jk} : j = 1,\dots,m\}$ of a_P^k on $\mathbb{C}^{2m}$, with ζ of order t, satisfies*
>
> $$A = d\cdot\mu_q - 2\cdot\{1\} \quad (d \text{ even}), \qquad A = d\cdot\mu_q - \{1\} - \{-1\} \quad (d \text{ odd}).$$

↳ Not used by another result of the paper.

Proof ▾

Since $t = 2m + 2$, the exponents $\{\pm j : j = 1, \dots, m\}$ are exactly $\mathbb{Z}/t \setminus \{0, t/2\}$. The map $e \mapsto \zeta^{ke}$ factors through $\mathbb{Z}/q$, because ζ^k has order q, and every residue mod q is the image of exactly d elements of $\mathbb{Z}/t$; hence $\{\zeta^{ke} : e \in \mathbb{Z}/t\} = d \cdot \mu_q$. It remains to delete the two excluded exponents. For $e = 0$ the element is 1. For $e = t/2 = dq/2$: if d is even this is $\equiv 0 \pmod q$ and the element is again 1; if d is odd then q is even, since $dq = t$ is, and $dq/2 \equiv q/2$, so the element is -1.

▾ 💡 **In plain words (commentary, not part of the paper)**

The exponents $\pm j$, $j = 1, \dots, m$, are all of $\mathbb{Z}/t$ except 0 and $t/2$, where $t = 2m + 2$. Raising to the k-th power maps $\mathbb{Z}/t$ onto $\mathbb{Z}/q$, d to one, so the whole of $\mathbb{Z}/t$ gives d copies of every q-th root of unity. Removing the two missing exponents: 0 gives 1, and $t/2$ gives 1 if d is even and -1 if d is odd. So the spectrum depends on d and q only, $\chi_\lambda(a_P^k) = \chi_\lambda(a_P^d)$, and one may take $p = 1$. The full multiset $d \cdot \mu_q$ is the alphabet of Prasad's character relationship; the paper says the two deleted letters change the occupancies of the fixed classes and produce the factor κ of Theorem B.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **No direct check.** `signo_Fq.py`, `valor_signo_recuento.py` and `dims_recuento.py` build the spectrum from this proposition, so they assume it. `formula_pura.py` and `valor_todo_q.py` evaluate at the eigenvalues of $g_{p/q}$ directly; their agreement with Theorem A is an indirect check.
- `valor_signo_recuento.py` V2: the spectra of a_P^k and a_P^d agree as multisets in 32 pairs ($m \le 5$), compared numerically with rounding, not in exact arithmetic.
- `cifras_alco.py` E5: $m = 6$, $k = 7$ gives $d = 7$, $q = 2$, eigenvalues 1 and -1 six times each.

Proposition 3.1. p. 5 📍 map

Let G be simple and simply connected, λ dominant, $\ell = \lambda + \rho$, $\theta = p/q$ in lowest terms, and $N_q(x) = \#\{\alpha > 0 : q \mid (x, \alpha)\}$, $r_q = N_q(\rho)$. Then $N_q(\ell) \ge r_q$, and

$$\chi_\lambda(g_{p/q}) \neq 0 \iff N_q(\ell) = r_q. \tag{1}$$

↳ Used in: Theorem A, Theorem C, Corollary 4.1, Theorem 9.1

Proof ▾

Each factor $1 - z^j$ of (2) has a simple zero at a primitive q-th root of unity exactly when $q \mid j$, so at $z = e^{2\pi i p/q}$ the numerator vanishes to order $N_q(\ell)$ and the denominator to order r_q. A character has no poles, so the first order is at least the second; and the limit is non-zero exactly when they agree.

▾ 💡 **In plain words (commentary, not part of the paper)**

Notation: $r_q = N_q(\rho)$. The statement holds in every simple simply connected group and at every rational point $g_{p/q}$, not only at the powers of a_P. In the product formula (equation (2) of the paper) each factor $1 - z^j$ has a simple zero at a primitive q-th root of unity exactly when $q \mid j$, so the numerator vanishes to order $N_q(\ell)$ and the denominator to order r_q. A character has no poles, so $N_q(\ell) \geq r_q$, and the limit is non-zero exactly when the two orders agree. When ρ is regular, $r_q = 0$ and this is the condition of Cellini–Möseneder Frajria–Papi; on the line of $\rho^\vee$, with coroots in place of roots, the same count is due to Nadimpalli–Pattanayak–Prasad, and the paper proves it by the same argument.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper. Programs check type C only; the other types are not checked.
- **Checked by program** `rho_line_any_q.py`: 27 183 dominant weights, $m = 2, \ldots, 6$, $q = 2, \ldots, 14$, $p = 1$, $\lambda_1 \leq 8$ ($m \leq 4$), ≤ 6 ($m = 5, 6$), box not stated in the paper; 0 failures, 19 986 of them with $q \nmid 2m + 2$ (`rho_line_any_q_OUT.txt`). **Floating point**: mpmath, 40 digits, χ counted as non-zero when $|\chi| \geq 10^{-6}$, 0 cases in the ambiguous zone; no rigorous error bound. Negative control: the naive rule ("some root value divisible by q") fails on 7968 weights.
- **Exact:** `formula_pura.py` checks the criterion in $\mathbb{Z}[X]/(\Phi_q)$ on every weight it visits, 0 failures: the whole box at the 45 pairs of the list of Theorem C, and up to the first counterexample of Theorem C at the other 158 pairs; and `comprueba_traducciones_aP.py` (a check in the blueprint's source, published with the blueprint, not part of the supplement) on 2336 cases with $q \mid 2m + 2$.
- `valor_todo_q.py`: 0 violations in the range of Theorem 10.1 (floating point).

Corollary 4.1. p. 6 map

> *Let $q \mid 2m + 2$ and $d = (2m + 2)/q$. Then $N_q(\ell) = r_q$ if and only if each folded class modulo q contains at most d of the ℓ_i, and each fixed class at most $\lfloor d/2 \rfloor$.*

↳ Used in: Theorem A, Theorem D, Proposition 5.1, Theorem 9.1

Proof ▾

By (5) the marginal cost of one more entry is $0, 1, 2, \ldots$ in a non-fixed class and $1, 3, 5, \ldots$ in a fixed one, so the places costing less than d number exactly d per non-fixed class and $\lfloor d/2 \rfloor$ per fixed one. There are $\lfloor q/2 \rfloor + 1$ folded classes, of which one is fixed when q is odd and two when q is even; since $m = dq/2 - 1$, the places of cost less than d number exactly m if d is odd and $m + 1$ if d is even. By Proposition 3.1 the minimum of N_q over the vectors $\ell = \lambda + \rho$ is r_q, and it is attained at ρ. It is also the minimum over all of $\mathbb{Z}^m$, hence over all placements of m entries in the folded classes:

every $x \in \mathbb{Z}^m$ is congruent modulo q to a vector $\lambda + \rho$ with λ dominant, namely $x_i + (m-i+1)Mq$ for a large integer M, whose entries are positive and strictly decreasing. If d is odd, reaching the minimum forces occupying all m cheap places, and any overflow replaces a place of cost $<d$ by one of cost $\geq d$. If d is even, the last admissible place of every class costs $d-1$, so omitting any one of them gives the same minimum, while exceeding a capacity again costs strictly more. So $N_q(\ell) = r_q$ if and only if no capacity is exceeded.

▾ 💡 **In plain words (commentary, not part of the paper)**

Equation (5) of the paper counts the vanishing factors class by class: $N_q(\ell) = \sum_{c \text{ not fixed}} \binom{n_c}{2} + \sum_{f \text{ fixed}} n_f^2$, with n_c the number of ℓ_i in the class. So one more entry costs $0, 1, 2, \ldots$ in a non-fixed class and $1, 3, 5, \ldots$ in a fixed one. The places costing less than d are d per non-fixed class and $\lfloor d/2 \rfloor$ per fixed class: m in all if d is odd, $m+1$ if d is even. The minimum of N_q is r_q, reached at ρ, and every placement of residues comes from some $\lambda + \rho$; so a character survives exactly when no class is over capacity. When d is odd every survivor has the occupancies of ρ; when d is even exactly one class is one entry short, which for ρ is the class 0.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Additional check, not part of the supplement:** `comprueba_traducciones_aP.py` (a check in the blueprint's source, published with the blueprint): the occupancy rule against $\chi \neq 0$, exact, 2336 cases, 0 failures; with the bound of the fixed classes changed to d it fails in 567 cases.
- `rho_line_any_q.py`, column D2: the capacity rule agrees with $\chi \neq 0$, computed numerically as for Proposition 3.1, on 7197 weights with $q \mid 2m+2$. The program labels it a negative control, but it is measured only where it must agree: a positive check of Theorem A(1) in its capacity form.
- `signo_Fq.py` O2: in all 10 230 surviving weights with d even, exactly one class is one entry short (the remark after the corollary).

3 The residue lemma

Proposition 5.1. p. 6 map

Work in $\mathbb{Z}[X]/(X^q-1)$. For $x = (x_1, \ldots, x_m) \in \mathbb{Z}^m$ put

$$P_x(X) = \sum_{i=1}^{m} \left(X^{x_i} + X^{-x_i}\right), \qquad \mathscr{R}_x(X) = \sum_{\alpha \in \Phi(C_m)} X^{(x,\alpha)}.$$

Then

$$\mathscr{R}_x(X) \;=\; \frac{P_x(X)^2 + P_x(X^2)}{2} \;-\; m, \qquad (2)$$

and, for $q \mid 2m+2$ and every λ satisfying the capacities of Corollary 4.1, $\mathscr{R}_{\lambda+\rho} = \mathscr{R}_\rho$.

↳ Used in: Theorem A, Theorem 10.1

Proof ▼

Expanding $P_x(X)^2$ collects $X^{\pm x_i \pm x_j}$ over all ordered pairs. The off-diagonal terms give each root $\pm e_i \pm e_j$ twice; the diagonal terms give $X^{\pm 2x_i}$ once each and $2m$ copies of X^0; and $P_x(X^2)$ supplies the second copy of each $X^{\pm 2x_i}$. Halving therefore leaves every root of C_m exactly once, together with m copies of X^0, and subtracting m removes those. That is (6).

For the second claim write $T_q(X) = 1 + X + \cdots + X^{q-1}$, so that $T_q(X)\,X^c = T_q(X)$ for every c. If d is odd the capacities are saturated, the folded occupancies of ℓ are those of ρ, and $P_\ell = P_\rho$ outright. If d is even, one class c falls a single unit below its capacity, and

$$P_\ell(X) = d\,T_q(X) - X^c - X^{-c}, \qquad P_\rho(X) = d\,T_q(X) - 2,$$

so that $P_\ell(X)^2 - P_\rho(X)^2 = X^{2c} + X^{-2c} - 2$ while $P_\ell(X^2) - P_\rho(X^2) = -X^{2c} - X^{-2c} + 2$. The two corrections cancel in (6), and $\mathscr{R}_\ell = \mathscr{R}_\rho$.

▾ 💡 **In plain words (commentary, not part of the paper)**

Notation: $P_x = \sum_i (X^{x_i} + X^{-x_i})$ in $\mathbb{Z}[X]/(X^q - 1)$, and $\mathcal{R}_x$ the sum of $X^{(x,\alpha)}$ over all roots of C_m. The identity $\mathcal{R}_x = (P_x^2 + P_x(X^2))/2 - m$ is the adjoint character minus the rank, since in type C the adjoint representation is the symmetric square of the natural one (Remark 5.2). The second claim: under the capacities of Corollary 4.1, $\mathcal{R}_\ell = \mathcal{R}_\rho$. For d odd the occupancies agree, so $P_\ell = P_\rho$. For d even one class c is one entry short; then $P_\ell^2 - P_\rho^2 = X^{2c} + X^{-2c} - 2$ and $P_\ell(X^2) - P_\rho(X^2)$ is its negative, so they cancel. In words: the root values of ℓ and of ρ coincide modulo q, up to sign, as multisets. This replaces Weyl conjugacy in the proof of Theorem A(2) (Example 6.1); outside type C it is the missing step (Question 12.2).

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `valor_signo_recuento.py` V1: the identity in 48 669 cases and $\mathcal{R}_\ell = \mathcal{R}_\rho$ on 11 972 surviving weights, $m = 2, \ldots, 7$, every k, $\lambda_1 \le 8$ ($m \le 4$), ≤ 6 ($m \ge 5$), box not stated in the paper; exact integer vectors, 0 failures (`valor_signo_recuento_OUT.txt`). No negative control (no mutation of the identity).

4 Every rational point

Proposition 8.1. p. 9 map

> *Let $m \geq 2$ and $\gcd(p,q)=1$. Then $\chi_\lambda(g_{1/q}) \in \mathbb{Z}$ for every dominant weight λ of $\mathrm{Sp}(2m)$ if and only if at least one of*
>
> $$q \mid 2m, \qquad q \mid 2m+1, \qquad q \mid 2m+2, \qquad q = 6 \text{ and } m \equiv 1 \pmod 3$$
>
> *holds; and in that case $g_{p/q}$ is conjugate to $g_{1/q}$ for every p coprime to q, so no character value depends on the numerator.*

↳ Used in: Theorem C, Theorem D

Proof ▾

Suppose every $\chi_\lambda(g_{1/q})$ is an integer. The exterior powers $\Lambda^j\mathbb{C}^{2m}$ are direct sums of irreducibles, so their characters are $\mathbb{Z}$-combinations of the χ_λ and are integers as well; those characters are the coefficients of the characteristic polynomial of $g_{1/q}$ on $\mathbb{C}^{2m}$, so that polynomial lies in $\mathbb{Z}[X]$, and the multiplicities of its roots are constant along Galois orbits. Conversely, if the spectrum is carried to itself by every $\xi_q \mapsto \xi_q^{\,p}$, then $g_{p/q}$ and $g_{1/q}$ have the same spectrum and are conjugate in $\mathrm{Sp}(2m)$; the character values are then algebraic integers fixed by the Galois group of $\mathbb{Q}(\xi_q)$, hence rational integers.

Write $m = uq + v$ with $0 \leq v < q$ and $a = \lfloor (q-1)/2 \rfloor$. The spectrum $\{\xi_q^{\pm j} : j = 1, \dots, m\}$ is, as a multiset of μ_q,

$$2u \cdot \mu_q \;+\; \{\pm 1, \dots, \pm v\},$$

and the first summand is Galois-stable. So the question is when the segment $\{\pm 1, \dots, \pm v\}$ is. It is stable for $v = 0$, $v = a$, $v = q-1$ and, when q is even, $v = q/2$. For necessity write $A_v = \sum_{j=1}^{v} (\xi_q^{\,j} + \xi_q^{\,-j})$; if the segment is stable then A_v is fixed by the whole Galois group. A full turn sums to zero, whence $A_v + A_{q-1-v} = -2$ and $\mathbb{Q}(A_v) = \mathbb{Q}(A_w)$ for $w = \min(v, q-1-v)$; the four segments just listed are $w = 0$ and $w = a$, so only $1 \leq w < a$ is left. There set $\beta_w = 1 + A_w = \sum_{j=-w}^{w} \xi_q^{\,j}$, whose conjugates are

$$\sigma_p(\beta_w) = \frac{\sin\big((2w+1)\pi p/q\big)}{\sin(\pi p/q)}.$$

For $q \geq 4$, $2 \leq N \leq q-2$ and $1 \leq r \leq q-1$ with $r \neq 1, q-1$, one has the strict inequality

$$\left| \frac{\sin(N\pi r/q)}{\sin(\pi r/q)} \right| < \frac{\sin(N\pi/q)}{\sin(\pi/q)}.$$

Indeed, replacing N by $n = \min(N, q-N)$ changes no modulus; on $0 < x < \pi/n$ the function $f(x) = \sin(nx)/\sin x$ is positive and strictly decreasing, since

$f'/f = n\cot(nx) - \cot x < 0$ because $t\cot t$ decreases on $(0,\pi)$; and on $\pi/n \le x \le \pi/2$ one has $|f(x)| \le 1/\sin(\pi/n) < 1/\sin(\pi/2n) = f(\pi/2n) \le f(\pi/q)$. Taking $N = 2w+1$, which lies in that range precisely when $1 \le w < a$, the only automorphisms fixing β_w are the identity and complex conjugation; so $\mathbb{Q}(\beta_w) = \mathbb{Q}(\xi_q + \xi_q^{-1})$, which is $\mathbb{Q}$ only when $\varphi(q) = 2$. Inside this range that leaves $q = 6$, $w = 1$, that is $(q,v) = (6,1)$ and $(6,4)$. Translating the surviving residues into divisibilities gives the four conditions.

▾ 💡 **In plain words (commentary, not part of the paper)**

When are all the characters at $g_{1/q}$ integers? If they are, the exterior powers show that the characteristic polynomial on $\mathbb{C}^{2m}$ has integer coefficients, so the spectrum is stable under Galois. Conversely a Galois-stable spectrum makes every $g_{p/q}$ conjugate to $g_{1/q}$, and the values are rational integers. Writing $m = uq + v$, the spectrum is $2u$ full turns of μ_q plus the segment $\{\pm 1, \dots, \pm v\}$, so everything rests on the segment. It is stable for $v = 0$, $v = a$ with $a = \lfloor (q-1)/2 \rfloor$, $v = q-1$ and, for q even, $v = q/2$. For the other v, the sum over the segment has a strictly largest conjugate, so it generates $\mathbb{Q}(\xi_q + \xi_q^{-1})$, which is $\mathbb{Q}$ only when $\varphi(q) = 2$; in that range this leaves $q = 6$. The key step is a lemma of Gannon–Schopieray; sufficiency for $q \mid 2m+1$ and $q \mid 2m+2$ also follows from Kac. Irrational θ never gives integers (Remark 8.2).

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Checked by program `galois.py`**: Galois stability of the spectrum against the four conditions on 11 800 pairs (m,q), $q \le 60$, $m \le 200$ (the range includes $m = 1$, outside the statement); exact combinatorics, 0 disagreements (`galois_OUT.txt`). This compares two conditions, not character values.
- Against the characters themselves, `galois.py` part (2): 44 pairs, 0 disagreements, **floating point** (mpmath, 40 digits, limit taken at a perturbed point, tolerance 10^{-12}, $\lambda_1 \le 5$).
- **Sufficiency, exact:** `prefactores.py`, integrality in $\mathbb{Z}[X]/(\Phi_q)$ on 52 928 pairs (λ, q) at the four families.
- Necessity against actual character values is checked only by the 44-pair floating-point run.

Theorem 9.1. p. 10 📍 **map**

$$S_{C_m}(q) = \begin{cases} \dfrac{m!\; 2^{\,2ua+v} \binom{a}{v}}{\left((2u+1)!\right)^{v} \left((2u)!\right)^{a-v} (u!)^{f}}, & v \le a, \\[2ex] \dfrac{m!\; 2^{\,a(2u+1)} \binom{a+f}{v-a}}{\left((2u+1)!\right)^{a} (u!)^{f} \,(u+1)^{\,v-a}}, & v > a. \end{cases}$$

↳ Used in: Corollary 10.2

Proof ▼

> By (5) the cost of an extra entry is $0, 1, 2, \ldots$ in a non-fixed class and $1, 3, 5, \ldots$ in a fixed one, so the minimum of N_q is attained by filling the cheapest places, and $\rho = (m, \ldots, 1)$ attains it, its residues being u full turns of $\mathbb{Z}/q$ followed by $1, \ldots, v$; by Proposition 3.1, and since every placement of entries in classes is the residue of some $\lambda + \rho$ (proof of Corollary 4.1), that minimum is r_q. When $v \leq a$, each fixed class carries u entries, v non-fixed classes carry $2u + 1$ and the remaining $a - v$ carry $2u$; the binomial $\binom{a}{v}$ chooses the v classes, the multinomial coefficient distributes the m coordinates, and each entry of a non-fixed class has two possible residues, which gives the power of 2. When $v > a$ one starts from $2u + 1$ in every non-fixed class and u in every fixed one and places the remaining $v - a$ entries in distinct classes, chosen in $\binom{a+f}{v-a}$ ways; adding an entry multiplies the count by $1/(u + 1)$ whether the class is fixed or not.

▾ 💡 **In plain words (commentary, not part of the paper)**

$S_X(q)$ counts the residues of weights modulo qP at which the criterion of Proposition 3.1 allows a non-zero character. Notation: $f = 1$ for q odd and 2 for q even (the number of fixed classes), $a = (q - f)/2$, $m = uq + v$. By the costs of Corollary 4.1, the minimum is reached by filling the cheapest places. When $v \leq a$: u entries in each fixed class, $2u + 1$ in v of the non-fixed classes, $2u$ in the rest. The binomial chooses the v classes, the multinomial distributes the coordinates, and each entry in a non-fixed class has two residues, hence the power of 2; $v > a$ is similar. Special values: $S_{C_m}(2m + 2) = 2^m m!$, $S_{C_3}(5) = 72$, $S_{C_3}(6) = 96$. It is not a quasi-polynomial: $S_{C_2}(q) = (q - 1)(q - 3)$ for large odd q, but $S_{C_2}(3) = 8$.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `valor_signo_recuento.py` V4: formula against exhaustive enumeration of $(\mathbb{Z}/q)^m$ on 40 pairs (m, q), $m \leq 4$, $q \leq 12$; exact, 0 discrepancies (`valor_signo_recuento_OUT.txt`).
- `cifras_alco.py` E6: $S_{C_3}(5) = 72$, $S_{C_3}(6) = 96$, $S_{C_2}(3) = 8$, and formula equal to enumeration on 36 pairs ($m \leq 4$, $q \leq 10$). `signo_Fq.py` O3 (the case $q > 2m$) and `dims_recuento.py` part (2) (the case $q \mid 2m + 2$) agree with a direct count. All exact.
- "$(q - 1)(q - 3)$ for every large odd q" follows from the formula; enumeration covers odd $q \leq 11$.
- No negative control.

Theorem 10.1. p. 11 📍 map

> *Let λ be a dominant weight of $\mathrm{Sp}(2m)$ with $N_q(\lambda + \rho) = r_q$. Then $b - \rho_Y$ is a dominant weight of Y, the element $g^Y_{p/q}$ is regular, and*

$$\chi_\lambda(g_{p/q}) \;=\; (-1)^{E_{p,q}(\lambda)}\,\kappa\,\Delta_\lambda\,\bigl|\chi^Y_{b-\rho_Y}\bigl(g^Y_{p/q}\bigr)\bigr|.$$

For $v \in \{0, a, q-1\}$, that is, for $q \mid m$, $q \mid 2m+1$ or $q \mid 2m+2$, the last factor is 1 for every such λ; for $q \mid 2m+2$ the theorem is Theorems A(2) and B.

↳ Used in: Theorem C, Corollary 10.2

Proof ▾

The class of ρ. The residues of ρ are u full turns of $\mathbb{Z}/q$ followed by $1, \dots, v$. If $v \le a$ the classes with $2u+1$ entries are $1, \dots, v$. If $v>a$ the classes with one entry fewer are 0 and the non-fixed $c<q-v$; the class $q/2$, for q even, is full, since $v \ge q/2$. In both cases these classes are the entries of ρ_Y.

Regularity. The entries of b are distinct and lie in $\{1, \dots, a\}$ if $Y = C_s$, and in $\{0, \dots, a\}$, together with $q/2$ when q is even, if $Y = D_s$. Hence every (b, α) with α a positive root of Y, namely $b_i \pm b_j$ and, for C_s, $2b_i$, lies strictly between 0 and q, since $2a<q$ and $a + q/2<q$ for q even. The same holds for ρ_Y, which is one of the possible b. So $b - \rho_Y$ is dominant and $g^Y_{p/q}$ is regular. The computation that gives (3) in Section 3 is stated there for G simple and simply connected, but it uses only the Weyl character formula and the Weyl denominator identity, with ρ a weight of the group; it therefore applies verbatim to $\mathrm{Sp}(2s)$ and to $Y = \mathrm{SO}(2s)$, where ρ_Y is a weight, including D_2, which is not simple, and D_1, a torus, where both sides are 1. It gives for Y

$$\bigl|\chi^Y_{b-\rho_Y}(g^Y_{p/q})\bigr| = \frac{\Pi^Y(b)}{\Pi^Y(\rho_Y)}, \qquad \Pi^Y(x) = \prod_{\alpha \in \Phi^+(Y)} \bigl|\sin\bigl(\pi p(x,\alpha)/q\bigr)\bigr|.$$

Residues. In $\mathbb{Z}[X]/(X^q-1)$, with P_x and T_q as in Proposition 5.1, a non-fixed class c holding n entries contributes $n(X^c + X^{-c})$ to P_ℓ, and a fixed class c contributes $2nX^c = n(X^c + X^{-c})$. Hence

$$P_\ell = 2u\,T_q + P_b \quad (v \le a), \qquad P_\ell = 2(u+1)\,T_q - P_b \quad (v>a),$$

and the same holds for ρ with ρ_Y in place of b. Since $T_q^2 = q\,T_q$ and $T_q\,P_x = 2s\,T_q$ for $x \in \mathbb{Z}^s$, every term of (6) that contains $T_q(X)$ or $T_q(X^2)$ is the same for ℓ and for ρ, and

$$\mathscr{R}_\ell - \mathscr{R}_\rho = \frac{P_b^2 \pm P_b(X^2)}{2} - \frac{P_{\rho_Y}^2 \pm P_{\rho_Y}(X^2)}{2} = \mathscr{R}^Y_b - \mathscr{R}^Y_{\rho_Y},$$

with the upper sign for $v \le a$. Here $\mathscr{R}^Y_x = \sum_{\alpha \in \Phi(Y)} X^{(x,\alpha)}$, which is $(P_x^2 + P_x(X^2))/2 - s$ for C_s by (6), and $(P_x^2 - P_x(X^2))/2 - s$ for D_s, whose roots $\pm e_i \pm e_j$ are those of C_s without the $\pm 2e_i$.

The value. The limit in the proof of Theorem A uses only $N_q(\ell) = r_q$, so

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)}\,\frac{Z_q(\ell)}{Z_q(\rho)}\cdot\frac{\Pi(\ell)}{\Pi(\rho)}, \qquad \Pi(x) = \prod_{\alpha>0,\ q\nmid(x,\alpha)} \bigl|\sin\bigl(\pi p(x,\alpha)/q\bigr)\bigr|.$$

Now $\Pi(x)^2$ is the product of $|\sin(\pi pc/q)|$ over the non-zero residues c of $\mathscr{R}_x$, counted with multiplicity, and $\Pi^Y(x)^2$ is the same product for $\mathscr{R}^Y_x$, none of whose residues is zero. The identity of residue multisets $\mathscr{R}_\ell + \mathscr{R}^Y_{\rho_Y} = \mathscr{R}_\rho + \mathscr{R}^Y_b$ therefore gives $\Pi(\ell)/\Pi(\rho) = \Pi^Y(b)/\Pi^Y(\rho_Y)$.

The dimensions. As in the proof of Theorem B, $Z_q(\ell)/Z_q(\rho)$ is Δ_λ times the ratio of the constants V_n, C_n, B_n for the occupancies of ℓ and of ρ, since the dimensions for ρ are 1 and the powers of q agree. If $v \le a$ the two multisets of occupancies agree class type by class type and the ratio is 1. If $v>a$, the occupancies agree outside the classes that fall short, a class falling short multiplies the constant by $V_{2u+1}/V_{2u+2} = 1/(2u+1)!$, by $B_u/B_{u+1} = 1/(2u+1)!$ or, for the class 0, by $C_u/C_{u+1} = 1/\bigl(2(2u+1)!\bigr)$, and both ℓ and ρ have $q-v$ such classes, the class 0 among them for ρ. The ratio is 2 if the class 0 is full for ℓ and 1 otherwise, that is, κ.

The cases $v \in \{0, a, q-1\}$. If $v=0$ then $s=0$ and $Y = C_0$. If $v=a$ then $Y = C_a$, and the a distinct entries of b in $\{1,\dots,a\}$ give $b = (a,\dots,1) = \rho_Y$. If $v = q-1$ then $Y = D_1$. In the three cases the last factor is 1. Now $v=0$ means $q \mid m$; $v=a$ means $2m+1 = (2u+1)q$ for q odd and $2m+2 = (2u+1)q$ for q even; and $v = q-1$ means $q \mid m+1$. For $q \mid 2m+2$, with $d = (2m+2)/q$, the case d odd is $v=a$ with q even, where $\kappa=1$, and the case d even is $v = q-1$, where $u+1 = d/2$; in both, κ is the $\kappa(\lambda)$ of Theorem B.

💡 In plain words (commentary, not part of the paper)

At a general rational point the residue multisets of ℓ and ρ differ, and the quotient of sines in the proof of Theorem A is no longer 1. This theorem identifies it. Let b list the classes holding $2u+1$ entries when $v \le a$, or, when $v>a$, the classes holding one entry fewer than the full occupancy ($2u+2$ for a non-fixed class, $u+1$ for a fixed one). Then $P_\ell = 2uT_q + P_b$ or $2(u+1)T_q - P_b$, every term containing T_q is common to ℓ and ρ, and $\mathcal{R}_\ell - \mathcal{R}_\rho = \mathcal{R}^Y_b - \mathcal{R}^Y_{\rho_Y}$ for a group $Y = C_s$ or D_s of rank $s \le m$ (equal to m only for $q>2m$, $Y = C_m$, and $q = 2m$, $Y = D_m$). So the quotient of sines is $|\chi^Y_{b-\rho_Y}(g^Y_{p/q})|$, a character of $\mathrm{Sp}(2s)$ or $\mathrm{SO}(2s)$ at a regular element, and $\chi_\lambda = \pm\kappa\,\Delta_\lambda\,|\chi^Y|$. For $v \in \{0, a, q-1\}$ the last factor is 1. Known cases: $v=0$ (Ayyer–Kumari), $q \mid 2m+1$ (Albion), $v=1$ (related to Kumari). The paper states that the reduction for $q \le 2m$, $2 \le v \le q-1$, $v \ne a$, and its residual factor of type D, are new.

🛡️ Evidence & scope

- **Not formalized in Lean.** Proof on paper.
- **Checked by program** `valor_todo_q.py`: all 4507 surviving (λ, p), $m = 2,\dots,5$, $q = 2,\dots,11$, p prime to q, $\lambda_1 \le 4$ (box stated in the paper); 0 failures (`valor_todo_q_OUT.txt`). **Floating point**: characters of $\mathrm{Sp}(2m)$,

$\mathrm{Sp}(2s)$, $\mathrm{SO}(2s)$ as Weyl determinant quotients at a perturbation of the element (mpmath, 700 digits), tolerance $10^{-6}(1+|\chi|)$; no rigorous error bound. The evaluator `carac.py` uses no formula of the paper; b, Y, κ, E are transcribed from the statement, which is what is tested. Negative control: $\rho_{D_s} \to (s, \dots, 1)$ breaks 1224 cases.

- `valor_todo_q.py casos`: 600 cases at six listed (m, q), among them 192 with $u = 1$, $Y = D_3$, $b \neq \rho_Y$; $\chi = \chi^{C_v}$ with $\Delta_\lambda = 1$ on 164 weights equal to their q-core with at most v parts. Same floating-point method; the control breaks 192.
- `prefactores.py` (e): 11 244 cases, $m \leq 6$, $q \leq 16$, 0 failures. Second evaluator, independent of `carac.py`: the character of $\mathrm{Sp}(2m)$ exact in `evlib.py`, then evaluated to 40 digits; χ^Y by the Weyl determinant; tolerance 10^{-25}. Floating point.
- `residual_factorizacion.py`: an earlier form of $|\chi|$, $p = 1$, 1814 weights, $m \leq 6$, $q \leq 12$, floating point; four mutations each break 355 to 390 of 1145 cases.
- **Gap:** away from the integer points of Proposition 8.1, every check of the residual factor for $v \notin \{0, a, q-1\}$ is floating point. At the integer points with $v \notin \{0, a, q-1\}$ ($q = 6$, $m \equiv 1, 4 \pmod 6$; $q \geq 4$ even, $q \mid 2m$, $q \nmid m$) it is checked exactly at $p = 1$, through $c_\lambda = \kappa R_1(b)$, by `prefactores.py` and `formula_pura.py`. The constant ratios $1/(2u+1)!$ and $1/(2(2u+1)!)$ have no program.

Corollary 10.2. p. 13 map

Let $q \mid m$ or $q \mid 2m+1$, let $\gcd(p, q) = 1$, and let λ be a dominant weight of $\mathrm{Sp}(2m)$. Then $\chi_\lambda(g_{p/q}) \neq 0$ if and only if each non-fixed folded class modulo q contains at most $2m/q$, respectively $(2m+1)/q$, of the entries of $\ell = \lambda + \rho$, and the fixed classes at most m/q, respectively $(2m+1-q)/(2q)$; these bounds are then attained. In that case

$$\chi_\lambda(g_{p/q}) \;=\; (-1)^{E_{p,q}(\lambda)}\,\frac{Z_q(\lambda+\rho)}{Z_q(\rho)}, \qquad |\chi_\lambda(g_{p/q})| \;=\; \Delta_\lambda,$$

where Δ_λ is the product of dimensions of Theorem B.

↳ Not used by another result of the paper.

Proof ▼

The two cases are $v = 0$, with $u = m/q$, and $v = a$ with q odd, with $u = (2m+1-q)/(2q)$. In both $v \leq a$, so the occupancies of the surviving weights, listed before Theorem 10.1, are $2u$ or $2u+1$ in every non-fixed class and u in every fixed class; they add up to m, so the bounds are attained, and a placement exceeding one of them costs more than r_q by (5) and the marginal costs in the proof of Theorem 9.1. Since $v \leq a$, $\kappa = 1$, and the last factor of Theorem 10.1 is 1. Its proof gives $\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)}\big(Z_q(\ell)/Z_q(\rho)\big)\,\Pi(\ell)/\Pi(\rho)$ with $\Pi(\ell)/\Pi(\rho)$ equal to that factor, and $Z_q(\ell)/Z_q(\rho) = \kappa\Delta_\lambda$.

▾ 💡 **In plain words (commentary, not part of the paper)**

The two families $q \mid m$ ($v = 0$) and $q \mid 2m+1$ ($v = a$, q odd) are cases of Theorem 10.1 where nothing is left over, as is $q \mid 2m+2$ with d odd: $v \le a$, so $\kappa = 1$, and the residual factor is 1. Survival becomes a pure capacity rule: at most $2m/q$, resp. $(2m+1)/q$, entries of ℓ in each non-fixed class and m/q, resp. $(2m+1-q)/(2q)$, in each fixed class, and the bounds are then attained. On the survivors the closed form of Theorem A(2) holds and $|\chi_\lambda| = \Delta_\lambda$. The paper attributes these values to known results: the case $v = 0$ is Ayyer–Kumari's factorisation at $X = (1, \dots, 1)$, and $q \mid 2m+1$ follows from Albion. The rest of the family $q \mid 2m$ behaves differently: at $\mathrm{Sp}(4)$, $q = 4$, $\lambda = (1,0)$ has Z-quotient 1 and value -2.

▾ 🛡️ **Evidence & scope**

- **Not formalized in Lean.** Proof on paper.
- **Exact:** `formula_pura.py` finds the closed form on every surviving (λ, p) of its box at each pair of these two families with $m \le 8$, $q \le 30$; `prefactores.py` finds $c = 1$, that is $|\chi_\lambda| = \Delta_\lambda$, in both families.
- `valor_todo_q.py casos`: 312 of 312 cases at $(m,q) = (3,3), (5,5), (2,5), (4,3), (6,13)$; floating point, as for Theorem 10.1.
- The capacity bounds are not checked separately; they are covered by the checks of Proposition 3.1.

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